

**ISSUE 3
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**MARLBOROUGH ACADEMIC
RESEARCH SOCIETY**

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SPOTLIGHT
INTERVIEWS**



M A R S

ISSUE III

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THE STAFF**

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MEET THE STAFF



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Our Mission

The Marlborough Academic Research Society (MARS) aims to provide Marlborough students from all backgrounds with the opportunity to edit, refine, and publish their research. By facilitating robust discussions across a diverse range of academic fields, and by teaching students how to navigate the peer review process, MARS hopes to prepare Marlborough students for future scholarly research and publication. MARS also looks to challenge and amplify existing knowledge through the dissemination of quality research in the sciences, social sciences, and humanities.

ISSUE III



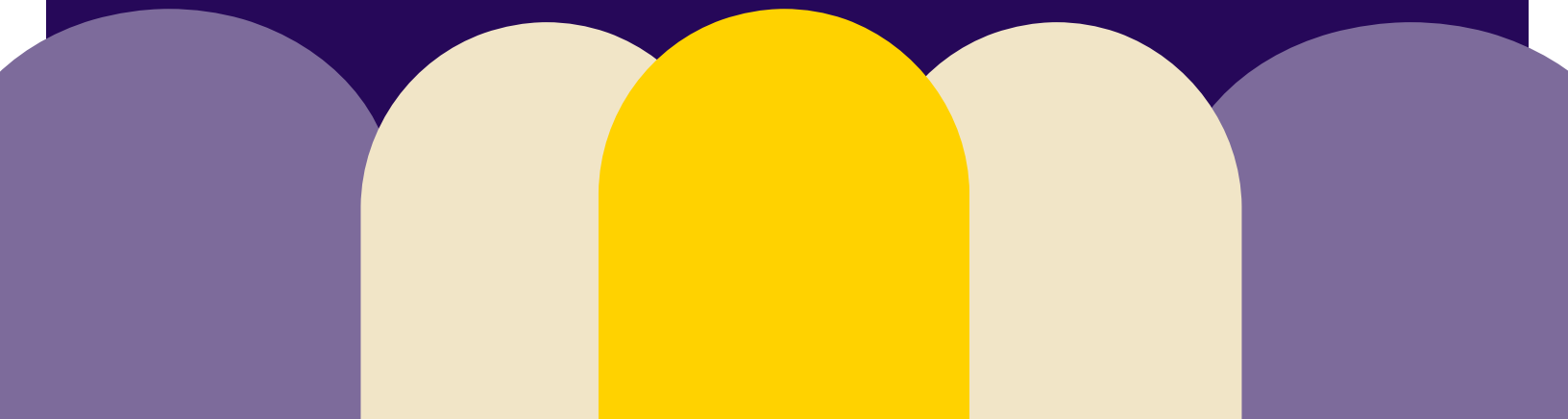
**This next section highlights the distinguished work and abstracts
of *some* of our reseachers!**

ABSTRACT #1

Computational Prediction of Antimicrobial Resistance in *Klebsiella Pneumoniae*

By Claire Lee

“Antimicrobial resistance (AMR) in *Klebsiella pneumoniae* poses a critical global health threat, especially with the emergence of carbapenem-resistant strains. This study used computational methods to predict antibiotic resistance patterns by analyzing 4,003 *K. pneumoniae* isolates from the BV-BRC database. Resistance genes were identified using ResFinder, and machine learning models were created to predict resistance across 20 antibiotics. SHAP (SHapley Additive exPlantations) analysis was used to determine which genes most strongly influenced predictions. The analysis showed high resistance rates to beta-lactam antibiotics, with over 80% resistance to ampicillin. Efflux pump genes *oqxA* and *oqxB* were present in over 90% of isolates, while beta-lactamase genes like *blaCTX-M-15* and *blaTEM-1B* were frequently detected. Machine learning models demonstrated substantially improved accuracy compared to rule-based prediction where XGBoost showed up to 21.6% improvement for certain antibiotics. SHAP analysis identified aminoglycoside-modifying enzymes (*armA*, *aac(6')-Ib*) as main predictors for amikacin resistance and carbapenemase genes (*blaKPC*, *blaNDM*, *blaOXA-48*) as more dominant predictors for carbapenem resistance. Overall, these findings indicate that computational approaches using machine learning can predict antimicrobial resistance patterns more accurately than gene presence alone, but limitations indicate that these tools should complement rather than replace traditional phenotypic testing.”

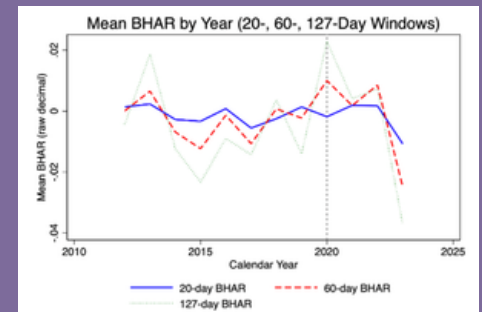


ABSTRACT #2

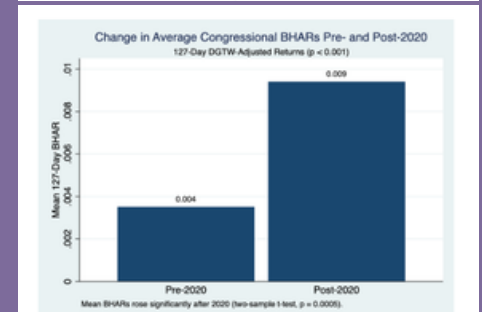
Taking Stock of Democracy: Abnormal Returns of High-Profile Representatives in Congress

By Leah Fischer

“United States Congressional insider trading has been the subject of much research and public attention in recent years. Observations of representatives with high net worths and frequent trading behavior have raised questions about the ethics and legality of Congressional stock trading. If members of Congress are permitted to make stock trades based on classified information with impunity, the interests of the people may become secondary to the desire for personal profit in determining US domestic and foreign policy. Despite scandal, economic researchers have found that the average member of Congress does not significantly outperform the stock market. In January of 2020, four Senators were publicly involved in insider trading scandals preceding the onset of the COVID-19 pandemic, but all investigations were quickly dropped by the Department of Justice. While insider trading research has continued in recent years, no studies have focused on the DOJ investigations as a turning point for insider trading or attempted to track the change in returns over time. In addition to exploring the aforementioned questions, this research isolated the traders who engage in the highest volume of stock trades each year and tracked their returns relative to Congress writ large. While Congress still does not outperform the market by a wide margin, this study found that mean 6-month buy-and-hold abnormal returns have significantly increased for both purchases and



This graph charts the average BHAR at the 20-, 60-, and 127-day benchmarks for each year in which data was collected. While fluctuations are not significant, returns have increased moderately over time.



The DGTW-adjusted 127-day BHAR has increased over time for Congress as a whole, with outperformance more than doubling from 0.4% to 0.9%. While results are highly statistically significant, the scale of outperformance is not enough to indicate systematized insider trading behavior.

sales in 2020-2023 relative to 2012-2020, likely as the deterrent effect of insider trading regulations has worn off. Additionally, the standard deviations of both of the aforementioned categories have grown over time, suggesting a general trend towards decreased risk aversion in Congressional stock picking.”

ABSTRACT #3

Animacy And Agency For Earth Beings: How Anishinaabe Writers Promote Powerful Climate Change Narratives

By Colette Wang

“As the climate crisis becomes an increasingly urgent global issue, we argue that it is crucial to consider alternative climate narratives to the dominant Western anthropocentric framework. In our paper, we study literature from the Anishinaabe peoples, a group of culturally and linguistically related Indigenous nations located in the Great Lakes region of North America. The primary objective of our research is to identify literary techniques that Anishinaabe writers employed to promote an inclusive narrative of the climate crisis. Through close reading and critical analysis, we examine how Anishinaabe writers portray “earth beings,” such as animals and plants, typically considered inferior by Western thinking. We argue that contemporary Anishinaabe authors employ literary techniques to advance themes of animacy and agency, offering an alternative understanding of the environment and interspecies relationships. By portraying earth beings as spiritual, autonomous, and powerful, Anishinaabe literary works further a system of thought that values all species equally. Our research advocates for the incorporation of literary methods used by Anishinaabe peoples so as to reframe Western perceptions of inanimacy.”

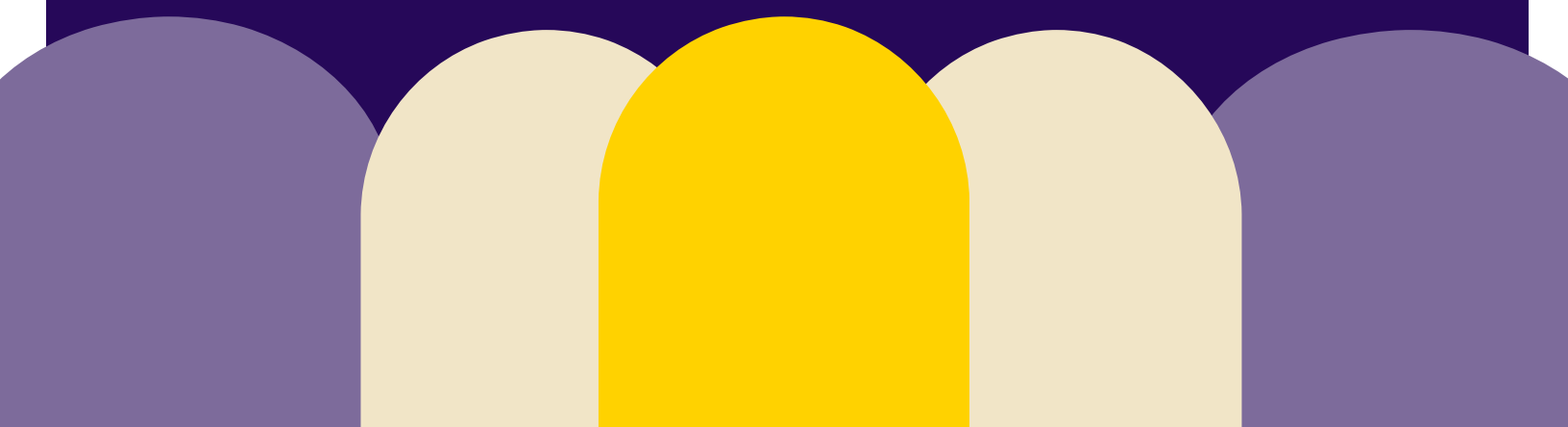
ABSTRACT #4

3-D Gait Analysis for Treatment Plans

By Elle Uhrman

“My research project delves into the advantages of 3D Motion Capture for Children with Cerebral Palsy. By using gait analysis, my study specifically analyzes children suffering from Equinus gait, plantarflexion of the ankle leading to toe walking, which can cause hip deformities, overall abnormalities, and pain. The patients go through the clinical trial twice, once before surgery (Gastrocnemius Recession and/or Tendo-Achilles Lengthening) and after surgery.

Following general anthropometric measurements and evaluations, the patient undergoes trials walking from one end of the room to the other while being attached with markers that create a digital simulation of their gait. The data is then compared to normal gait as well as themselves pre-surgery and post-surgery using different software programs to visualize and quantify improvements in capabilities. Overall, my study aims to see the efficacy in predicting improvements or suggesting best recovery, as well as clearly highlighting said improvements post-surgery.”



STUDENT SPOTLIGHT

Through interviews with Honors Research students, this section reveals the inspiration, memories, and fun behind projects!

RACHEL CHEN

Representation Theory

What Are You Researching?

My research deals with the mathematical object called the spin representation in a field that is known as representation theory. Particles have many intrinsic properties, like mass and charge. One of these intrinsic properties is spin which describes the interaction of these particles with magnetic fields. A single electron is simple to think about; however, if you have many electrons in a system, these particles interact with each other and become complicated. This system is called the spin representation. I show how we can describe this spin representation with another mathematical object called the Temperley–Lieb algebra, which is an object that can be very easily visualized through a sequence of dots and lines.

Day-to-Day Responsibilities:

I don't really go to the lab; I do research at home. Sometimes I meet with my mentor and we discuss ideas. Sometimes, I refine my paper, and sometimes I just write on paper trying to understand my research!

Research Highlights:

All of my findings are really cool! I have had the opportunity to present at a conference at MIT. Additionally, I went to Washington D.C. in January and presented at another conference. Lastly, I am going to finals week for the Regeneron Science Talent research, which is a competition where I get to meet some other really cool people.

Challenges:

When things don't work out and I don't understand why things don't work out. For example, I thought that the canonical basis of spherical models looked like something different from what it obviously does. My mentor was laughing at me for a while.

Advice for Others:

It's fun! For math research, always start with learning. Learn as much math as you like; learn the math that you are interested in. It slowly turns into research.

SOPHIA BAAS-MAYNARD

MicroED-Guided Structural Elucidation of Secondary Metabolites from Natural Sources

What is your research topic?

- My research topic is MicroED-Guided Structural Elucidation of Secondary Metabolites from Natural Sources. What I do in very simple terms is I take plants, extract out the natural compounds, crystallize them, put them in a transmission electron microscope (TEM) and shoot electrons at the crystals, collecting movies while doing so. I then take the movies and process the data, which is then put in Olex2 to elucidate the 3D crystal structure in order to refine the structure. After determining the structure, the compound can be tested for antimicrobial properties which is helpful for growing antimicrobial resistance (AMR).

What inspired you to pursue this topic?

- My mom is a chemistry professor at UCLA, and she went to Caltech for graduate school. She would always talk to me about how great her experience was there, describing Caltech as a place that makes you want to research all of the time since everyone there is super passionate about science. After working last summer at UCLA under Professor Abigail Doyle I was so inspired that I wanted to continue researching and to explore a new area of chemistry. When thinking about what topic I could focus on, I found Professor Hosea Nelson and became interested in MicroED. Specifically, I became interested in different plants and looking up their secondary metabolites, which led me to look for newly discovered plants.

What impact do you hope your research will have?

- I hope that I will be able to discover a new compound that has antimicrobial properties. I also hope to inspire younger generations that it is possible to do high-impact research at a young age. Just because we don't have degrees doesn't mean our work isn't valuable.

What has been your favorite part of the research process?

- My favorite part of the research process is when I am having trouble with the crystallization process and collecting good movies, yet after recrystallization I can collect a really good movie. Seeing the electrons at high resolution/low angstrom makes me really excited.

A memorable moment (challenge, breakthrough, or unexpected discovery).

- One memorable moment was when I presented at the national meeting for the American Chemical Society (ACS) in Washington DC. I was able to discuss with professors, undergraduates and graduates in the same field as me. This experience was transformative as I got to see how my work was impactful and important in the wider chemistry field.

SOPHIA BAAS-MAYNARD

MicroED-Guided Structural Elucidation of Secondary Metabolites from Natural Sources

One research tip you would give to students just starting out.

- Don't be scared to talk to your mentor because they want to help you. Also, make sure you are awake and focused when you get to the lab so that you don't make a mistake that could damage equipment or ruin experiments. Communicating with your mentor will show them how responsible you are.

What skills has research helped you develop?

- Scientific writing is very different from writing an essay, and research has helped me practice this for my future career.

How long have you been researching?

- My first research project started in July of 2024 at UCLA. From that internship, I was able to get my research published in the Journal of the American Chemical Society, and present at the ACS national meeting. My second research project started June of 2025 at Caltech and it is currently ongoing. My future research will start in June of 2026 at the University of Groningen.

Where do you usually do your research?

- Caltech in the synthesis lab or in the MicroED lab.

If you have a routine, could you describe it?

- Over the summer I would get to Caltech at 8:30 AM and start by running the high performance liquid chromatography (HPLC) instrument, then I worked on the TEM or on crystallization. After collecting on the TEM I would go to lunch with the other high school students at Brown Cafeteria at 1 PM (sushi and dp strawberry). At 1:45 PM I will continue researching. In the afternoon, I would often have more free time to do other work or relax. I would finish up my work and go home anywhere between 6PM and 9 PM.

Coffee, tea, or something else while working?

- I love strawberry Dr Pepper, but you can't have it in the lab that's a no-no!

Any tips / tricks that have been helpful for you?

- You can have a notebook LM make a podcast of relevant research papers for you that you can listen to when driving to the lab. Start planning out writing your research paper a month in advance so the work doesn't pile up on you.

One to two words to describe research:

- Trailblazing and possibility

ELLE UHRMAN

3-D Gait Analysis for Treatment Plans

What is your research topic?

My research topic is the efficacy of 3-D Motion Capture Gait Analysis for Children With Cerebral Palsy in Determining Treatment Plans for Gait Improvements.

What inspired you to pursue this topic?

I've always been passionate about movement disorders ever since my grandpa had multiple sclerosis. I also really enjoy learning about things I'm not familiar with and I didn't know much about cerebral palsy before joining my lab.

What impact do you hope your research will have?

I hope my research allows for better treatment plans for kids with cerebral palsy because surgery is a very important step in treatment and it's important that kids get the best tailored treatment because cerebral palsy is an umbrella term for multiple different diverse conditions.

What has been your favorite part of the research process?

My favorite part is learning new skills, especially computer science and observing clinical tests. A research tip I would give to new students is to be incredibly open minded. Lean into the fact that you know nothing as that is the best way to broaden your knowledge of the world and empathy for others.

Where do you usually do your research?

I do my research at the Children's Hospital of Los Angeles in the Orthopedic Motion Lab. I process patient files into readable data using visual 3-D.

If you have a routine, could you describe it?

If I'm doing a lot of files, I like listening to music and talking with the physical therapist, Mia.



What is your research topic?

My research topic is the intersection of AI and education specifically regarding using AI in peer tutoring. I am really interested in specifically students' learning experiences, measures of students' academic self-efficacy (confidence and perceived math aptitude), peer tutors' perception of their teaching abilities, and the tutors' and tutees' perception of Artificial Intelligence over time.

What was your inspiration for your research question/topic?

I remember there was a classmate of mine who was speaking about going into a tutoring session and she was telling me, "I forgot everything. I have no idea how I am going to help the girl I am tutoring." That is the moment I realized that action had to be taken and changes had to be made to help peer tutors feel more ready to tutor others. I realize that Marlborough does not provide much training to peer tutors before allowing them to tutor individuals. Students also are so busy with other commitments and so students lack time for tutoring training. Therefore to provide a solution, I looked to the power of Artificial Intelligence, not as a threat to take over tutoring jobs, but as a tool to help peer tutors feel supported during sessions.

What do you hope to learn/accomplish through your research?

I hope to display that AI, while it may seem scary or completely negative, there can be positive outcomes to using it. I have been interested in the ways AI could be used in education for a while because education has always been an integral part of my life. However, while I have been blessed with a quality education, I understand that many people do not have the same privilege that I do. For some individuals, it is difficult to access quality tutoring because of the high cost and the shortage of quality tutors. Therefore, I see peer-tutoring as a potential outlet to provide quality tutoring without the high price involved. Through AI, I believe there is a possibility to minimize the education gap by maintaining the humanity in tutoring while enhancing the expertise of the tutors. Consequently, I hope to show that AI-assisted peer-tutoring could be a way to continue to have human connection while having a more beneficial tutoring experience.

What was your favorite part of your research process?

My favorite part of my research process was finalizing my research idea and finding a question that has never been researched before. I knew the main idea of what I wanted to study early on, right when I found out I was accepted into honors research which was at the end of last year. Then at the beginning of this year, I finalized exactly what I wanted to research, finding something that no one has investigated before. While I saw that individuals such as a researcher from Stanford had studied the tutors' perspective on AI-assisted tutoring, there was minimal insight from tutees.

Have there been any challenges or unexpected moments during your journey? If not, something you have learned?

Initially my entire research was going to be qualitative focused. However, I realized that none of my data would be statistically significant because my control size is only 8 people which meant that any quantitative data would only be applicable to those individuals. At first, I was terrified that my entire research project would fall apart. So to fix my issue, I decided to switch gears and took on a more qualitative approach, changing my research question and deciding to interview both the tutors and tutees. With qualitative data, you can understand every individual's perspective in depth and really understand nuanced outlooks on the situation rather than relying on patterns to generalize findings. In a lot of ways, qualitative data is more meaningful than quantitative data and now I am happy with my decision to switch.

Any tips for people starting out or wanting to get into research?

Research has a lot of ups and downs. It may be scary, but you have to be prepared to make changes that are sometimes drastic to your project and shape it into something you did not think it would turn out to be. In the end, it is all going to work out as it should, changes are part of the learning process. :)

How is managing time with everything else? When do you usually do your research?

My free period has been so important to honors research. Almost every one of my free periods is spent meeting with Mx. Tuluenga or working on my research paper. Therefore, I can spend time throughout the school getting my research hours in. Additionally, anything related to my research project that I do counts as hours. This means listening to an AI in education podcast for example or watching an AI in education video does contribute to my research hours which I find very helpful. Therefore I do not have to feel much pressure or worry that I won't be able to complete all my hours because these are fun ways to get them in.

What is your usual research routine?

I was lucky enough to find my mentor at an AI related event that I went to, so that was my first step. I think the most important first step, however, is to find the question that I wanted to study. To do so, I looked into Science Direct, Google Scholar, and JSTOR to find the gaps in my research topic and then decided my research question from there. With these sources I made summaries of them and put them into my bibliography to make it easier to write my lit review. Then I would email a mentor and tell them about the overall idea I would be thinking of, generate my materials, and get started.

EMMALINE EDWARDS

Umbrellas in a Hurricane: "Operation Amsterdam" and the 1944 Slovak Uprising

What is your research topic?

- I am currently brainstorming a title, but my research paper is about the many characters of the Slovak Uprising of 1944, from a grand scale to the individual. My long-term goal is to translate my findings into a spec script for a television show (likened to Band of Brothers) with the goal of maintaining as much factual accuracy as possible. This goal might be started next year in Capstone, or college if I choose to keep going on my research next year.

What inspired you to pursue this topic?

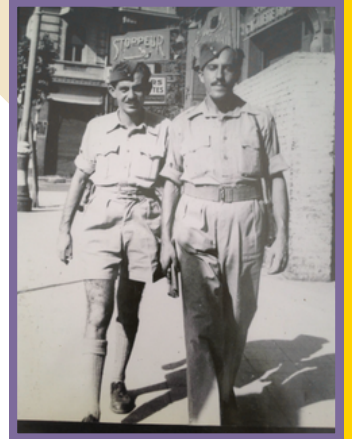
- For all my life, I have mourned the fact that I did not have any stories about my family as most evidence of their existence was destroyed in the Nazi strategy to erase Jewish stories and people. I was able to connect with an uncle, the namesake of the man involved in WWII and the Slovak Uprising of 1944. Rafael "Rafi" Reiss volunteered to serve the British and partisans to help defeat the Nazis and trained as a spy, diplomat, footsoldier, and parachutist. I am using the survivor Haim Hermesh's memoir "Operation Amsterdam" (translated into English by my great-uncle Raphael Reisz) to decipher the motives and events of the Uprising.

What impact do you hope your research will have?

- I hope to educate myself and others on the events that followed and help to memorialize the names of those involved. I also hope that, in doing this research, I can get to know my great-great-uncle posthumously.

What has been your favorite part of the research process?

- My favorite part of this research project has been learning of all the smaller stories within the bigger narrative. I've been introduced to wildly unique people and have enjoyed discussions with my family about this subject.



EMMALINE EDWARDS

Umbrellas in a Hurricane: "Operation Amsterdam" and the 1944 Slovak Uprising

A memorable moment?

- One of my favorite stories that I've come across in my research has been that of Albert the Spy, a "gentile" Frenchman described as "common" in the book. Albert quickly became known within the ranks as one of the most famous spies due to his repeated capture and escape from SS prisons. When the Allies needed someone to be rescued from an SS prison or wanted a message to get to a POW, they called upon Albert to get himself captured and relay the message to whomever needed it. Albert escaped different SS prisons SEVEN TIMES.

How long have you been researching?

- I have been researching this since last year, but began substantial research over the summer. After being paired with my mentor Gerald Steinacher, I was able to research with more structure.

Where do you usually do your research?

- I generally do research and write at my desk in my room and have reached out to the British Archives and Slovak Museum of the Partisans (SNP) via email. I have gotten no response yet but am still hopeful.

If you have a routine, could you describe it?

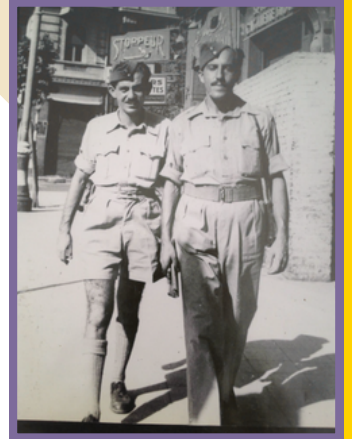
- I don't tend to do well with routines, so I just sort of allot a time in the evening for me to work and research whatever tickles my fancy relating to my project.

Coffee, tea, or something else while working?

- I am a diehard tea drinker and love a brew of earl grey when researching in the morning/afternoon, and rooibos in the evening.

One word to describe your research:

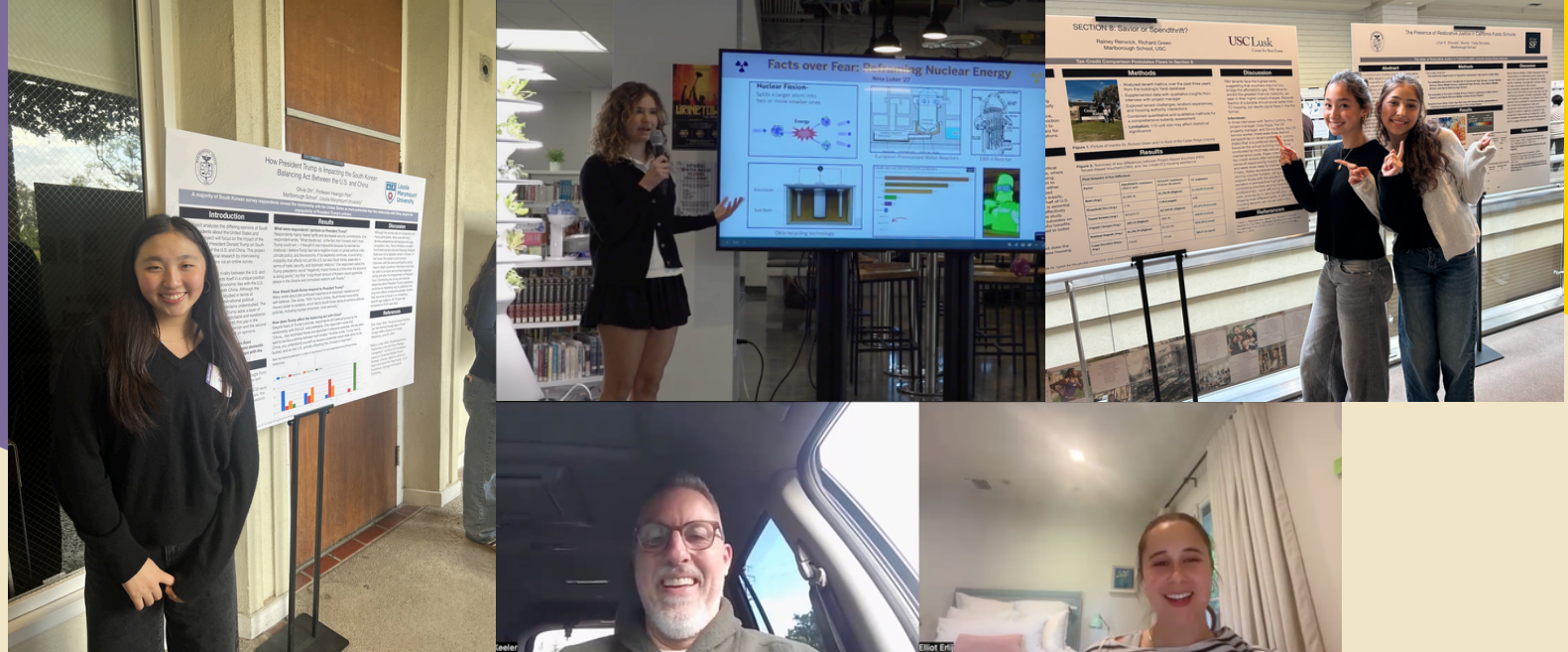
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25-26 IN PHOTOS



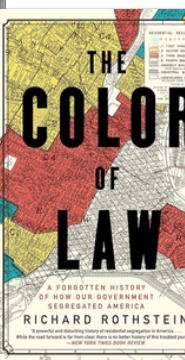
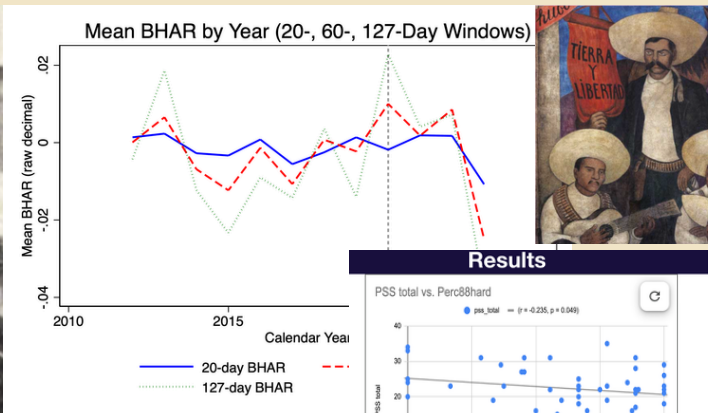
Keep reading to see photos and primary sources of our research
this year!



COI AND MENTOR FUN!



PRESENTATIONS



...AND
PRIMARY
SOURCES!

Scan to see **full papers and more!**



Thank you to Dr. Ponzio, Dr. Lynch,
and Mx. Tuluenga for your *invaluable*
support of MARS. Both our research
and publication would not be possible
without your mentorship!